DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

ENTERPRISE INFORMATION SYSTEMS

Question Bank with Model Answers

Course Title	Enterprise Information Systems	**Course Code**	IM355TBB
Semester	V	**Units Covered**	Unit I and Unit II
Compiled From	CIE and SEE papers, 2024–2026	**Questions**	Part A: 28 Part B: 28

Notes:

1. Questions are compiled from the CIE and SEE question papers of 2024–2026. The unit and the source paper are shown below each question.
2. Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive, application and case-based questions (8 to 10 marks).
3. Model answers are written to the depth expected for the marks allotted. Numerical answers show the complete working, and the final result is stated after **Answer:**.
4. Where a question repeats an earlier one, the answer refers to the earlier question (the reference is a link).
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome. Case-based questions are answered with a structured analysis; examples given are illustrative.

Sl. No.	Questions	M	BT	CO
	PART – A			
1	List two key features of a Customer Relationship Management (CRM) system. Source: Unit I | CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q1.1	02	2	1

- **Contact and lead management:** a single customer database recording every interaction, with lead generation, tracking and follow-up (sales force automation).
- **Customer service and analytics:** service-request handling, campaign management and analysis of buying patterns for segmentation and loyalty programmes.

2	Mention two types of data commonly handled by EIS. Source: Unit I | CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q1.2	02	2	1

- **Transactional (structured) data:** sales orders, purchase orders, invoices, payroll and inventory records stored in relational databases.
- **Master and unstructured data:** customer, supplier and material master records, and unstructured data such as e-mails, documents, XML files and web content.

3	Name two popular ERP software solutions. Source: Unit I | CIE-I, 27/11/2024	02	1	2

SAP S/4HANA (SAP AG, Germany) and **Oracle ERP Cloud / Oracle E-Business Suite**. Others: Microsoft Dynamics 365, Infor, Epicor, Ramco (India), Tally Prime for small firms.

4	What is the primary function of Supply Chain Management (SCM) within EIS? Source: Unit I | CIE-I, 27/11/2024	02	1	1

SCM within EIS coordinates the flow of materials, information and finances from suppliers to customers. Its primary function is to deliver the right product, in the right quantity, at the right time and at the right cost, through integrated planning, procurement, inventory, production and distribution.

5	What are the core modules of an ERP system? Source: Unit I | CIE-I, 27/11/2024	02	2	2

Financial accounting (FI) and controlling (CO), human resource management (HR), materials management (MM: purchasing and inventory), production planning (PP), sales and distribution (SD), plant maintenance (PM), quality management (QM), project systems (PS) and business intelligence/reporting.

6	_____ characteristic is typically associated with an Enterprise Information System (EIS). Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q1.1	01	1	1

Answer: integrated (enterprise-wide integration through a single common database)

7	_____ is the main benefit of EIS from a technical standpoint. Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q1.2	01	2	2

Answer: Data integration: data entered once into a single common database and shared by all functions

8	List two fundamental ways an Enterprise Information System contributes to improving an organization's operational efficiency. Source: Unit I | SEE Feb/Mar 2026, Q1.1	02	1	1

- **Integration and elimination of duplication:** data entered once at the most accurate source is shared by all departments, removing re-entry, errors and reconciliation effort.
- **Automation of routine transactions and real-time information:** e.g. automatic purchase orders when stock falls below the reorder point, and dashboards for faster operational decisions.

9	Enterprise Systems are noted for providing a tightly integrated solution to an organization's information needs. State two functions that were historically stand-alone applications. Source: Unit I | CIE-I, 10/11/2025	02	2	2

Stand-alone applications that were integrated in an enterprise system include **inventory control**, **sales and receivables**, **financial accounting** and **payroll/human resources** (any two).

10	The primary mechanism by which Enterprise Information Technology provides value to organizations is through appropriate _____ improvement. Source: Unit I | CIE-I, 10/11/2025	01	1	1

Answer: process (business-process) improvement

11	What is the role of EIS in business decision-making? State one key function. Source: Unit I | CIE-I, 10/11/2025 (Q4 and Q6)	02	1	1

EIS integrates enterprise data and processes into one framework and gives managers at all levels timely, accurate information for informed decisions. **One key function:** enhancing information availability: real-time, enterprise-wide data (e.g. an ERP dashboard showing current sales, inventory and production) for decision-making. (Others: strategic planning support, data-driven analytics, real-time monitoring of KPIs.)

12	State how data from an Enterprise Information System (EIS) supports strategic-level decision-making. Source: Unit I | SEE Feb/Mar 2026, Q1.6	02	2	2

Integrated data from ERP, CRM and SCM modules is summarized into forecasts, trend analyses and executive dashboards. Top management uses these insights into market trends, customer behaviour and supply-chain performance for long-term decisions such as capacity expansion, investment, new markets and product strategy.

13	How does the implementation of an integrated EIS/ERP system (like the one used in the Sanyo case) achieve “process quality improvement”? Source: Unit I | SEE Feb/Mar 2026, Q1.8	02	1	2

An integrated EIS/ERP improves process quality by replacing fragmented, manual processes with standardized best-practice processes on a single database: data are entered once and checked, errors and duplicate entries are eliminated, every transaction is traceable, and real-time information allows deviations to be detected and corrected quickly.

14	A textile company is implementing a new EIS. Its goal is to maximize the utilization of collective employee knowledge for project teaming and innovation. Which core EIS component is primarily responsible for managing and exploiting this intellectual capital? Justify your choice. Source: Unit I | CIE-I, 10/11/2025	02	3	3

Knowledge Management (knowledge work systems / business-intelligence portal). It captures, stores and shares the organization’s intellectual capital (expertise, documents, project experience) and identifies employees with relevant competence for project teams. Example: Ericsson’s Business Intelligence Centre portal, which profiled users automatically and helped create strong project teams by identifying employees with relevant internal competence.

15	Define the term 'Digital Transformation' (DT). Source: Unit I | SEE Feb/Mar 2026, Q1.2	02	1	1

Digital transformation is the strategic integration of digital technologies (cloud computing, AI, IoT, data analytics) into all areas of an organization to fundamentally change how it operates and delivers value to customers. It goes beyond digitization: it re-engineers processes, culture and customer experience.

16	The evolution of planning systems progressed from MRP-I to _____ and then to ERP. Source: Unit I | CIE-I, 10/11/2025	01	1	1

Answer: MRP-II (Manufacturing Resource Planning)

17	What is the primary purpose of Material Requirements Planning (MRP)? Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q1.3	01	1	2

MRP determines what, how much and when to order or produce: it translates the master production schedule for end items into time-phased net requirements for sub-assemblies, components and raw materials, so that they arrive just when needed (solving the problem of dependent demand) while keeping inventory low.

18	What does MRP-3 (Money Resource Planning) primarily focus on? Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q1.16	01	2	1

MRP-III (Money Resource Planning, or ERP-II) emphasizes the **planning of money (capital)** in an optimal manner: planning of capital and managing the situation when surplus money arises.

19	List a common challenge faced during ERP implementation related to software customization. Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q1.4	02	2	1

Heavy customization of the vendor's software to fit existing practices adds time and cost to implementation, makes upgrades difficult and dilutes the integration benefits of the ERP (less ability to communicate seamlessly with other modules, suppliers and customers).

20	Define Business Process Automation (BPA). Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q1.3	02	2	1

Business Process Automation (BPA) is the use of technology (workflow software, integration, rules engines, RPA) to execute recurring, multi-step business processes with minimal human intervention, so that the process runs faster, more consistently and at lower cost. Example: automated purchase-requisition approval and purchase-order generation.

21	Mention one advantage of implementing automation in business processes. Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q1.4	02	1	1

Reduced processing time and cost with fewer errors: repetitive tasks are executed faster, accurately and round the clock, freeing employees for higher-value work (also: consistency, audit trail, scalability).

22	Define Robotic Process Automation (RPA) and state its primary focus. Source: Unit II | CIE-II, 17/12/2025; SEE Feb/Mar 2026, Q1.3	02	1	1

RPA is a technology in which software robots (bots) mimic the actions of a human user on computer applications (logging in, copying data, filling forms, clicking buttons) to execute rule-based tasks. **Primary focus:** automating high-volume, repetitive, rule-based tasks on structured data across existing applications, without changing the underlying systems.

23	Name two common tools (leading commercial platforms) used for implementing RPA. Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q1.9; SEE Feb/Mar 2026, Q1.4	02	2	1

UiPath and **Automation Anywhere** (also Blue Prism and Microsoft Power Automate).

24	State the fundamental reason why a company must engage in Business Process Reengineering (BPR) before implementing any major automation initiative. Source: Unit II | CIE-II, 17/12/2025	02	2	2

Automating a poorly designed process only makes the inefficiency faster ("paving the cow path"). BPR first removes non-value-adding steps, redundancies and hand-offs and redesigns the process around outcomes, so that automation is applied to an efficient, standardized process and delivers its full benefit.

25	What is one of the risks associated with Business Process Reengineering? Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q1.5	01	2	2

High risk of failure due to employee resistance to radical change, often combined with layoffs and large investments in IT; many BPR projects fail to deliver the expected results.

26	_____ level of process quality design focuses on optimizing individual tasks within a process. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q1.6	01	1	1

Answer: Task (activity) level

27	Write the full form of BPM. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q1.7	01	2	1

Answer: Business Process Management

28	How does reengineering differ from incremental improvements? Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q1.8	01	1	2

Reengineering is a fundamental, radical redesign of a process from a clean slate for dramatic gains, whereas incremental improvement makes small, continuous changes to the existing process.

Sl. No.	Questions	M	BT	CO
	PART – B			
1	Analyse the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answer with examples. Source: Unit I | CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q2a	10	2	1

An Enterprise Information System is an integrated information system that uses a single common database to support all functions of an enterprise (finance, materials, production, sales, HR) and its links with suppliers and customers.

Benefits in a manufacturing organization

(1) **System integration:** an integrated system shared by all users replaces a diverse set of applications that rarely communicate; sub-organizational silos are eliminated. **Example:** a sales order automatically triggers production planning, purchasing, warehousing, shipping and accounts receivable (the order-to-payment process).
(2) **Data integration and accuracy:** data are entered once at the most accurate source and shared by all, giving more complete and accurate data.
(3) **Best business practices:** the package embeds proven processes, improving operational quality and productivity.
(4) **Better planning and inventory control:** MRP/MRP-II functions plan materials and capacity, reduce inventory and lead times and improve on-time delivery.
(5) **Improved decision-making:** real-time dashboards on production, inventory and cost support managers at all levels.
(6) **Lower cost and supply-chain linkage:** lower IT costs in the long run, fewer manual processes, and on-line connectivity with suppliers and customers (e.g. the State of South Carolina's pilot saved about $600,000 and eliminated a one-week procurement backlog).

Challenges

(1) **Less flexibility:** the system imposes one procedure on the whole organization, and everyone must conform.
(2) **High cost and time:** licences, hardware, consultants and training; full systems may cost from $5 million to over $100 million, and training is usually under-budgeted.
(3) **Change management:** people who have followed the same practices for years resist change; cultural issues are often a bigger hurdle than technology.
(4) **Customization issues:** modifying vendor software increases cost and reduces integration benefits.
(5) **Data migration and errors:** corrections are harder to make because controls are needed to ensure changes do not introduce new errors.
(6) **Hidden costs:** lost productivity while employees cope with the new system; dependence on vendor and consultants.

Conclusion: the benefits of integration are usually much greater than the costs of conformity, provided top management commits to the project, processes are reengineered first and users are trained.

2	How can information technology support a company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answer. Source: Unit I | CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q2b	10	3	3

Information technology (hardware, software, networks and databases used to manage and deliver information) supports a company in three ways:

1. Supporting business processes

- **Automation of transactions:** transaction processing systems record orders, payments and payroll quickly and accurately (e.g. airline and IRCTC reservations, ATM banking).
- **Integration of processes:** ERP links finance, HR, manufacturing and warehouse so that someone in finance can see whether an order has been shipped.
- **Collaboration:** e-mail, workflow and intranet portals connect employees, suppliers and customers (e.g. EDI for purchasing, avoiding paper purchase orders and invoices).

2. Supporting decision-making

- **MIS** provide routine reports for middle managers; **DSS** bring significant data to a manager's notice for semi-structured decisions; **ESS** give top management summarized trends through graphics and dashboards.
- **Business intelligence and analytics** detect trends and anomalies (e.g. dashboards highlighting underperforming products or regions).

3. Creating competitive advantage

- **Cost leadership:** automation and better inventory control reduce costs (e.g. Walmart's data sharing with suppliers).
- **Differentiation and customer intimacy:** CRM data enable personalized service and loyalty programmes (e.g. Amazon's recommendations).
- **Speed and agility:** faster response to market changes, shorter product-development cycles (e.g. Zara's quick design-to-store cycle).
- **Knowledge exploitation:** Ericsson's Business Intelligence Centre portal raised productivity by making internal knowledge and experts easy to find.

3	Define and differentiate the core components of EIS. Briefly define ERP, CRM and SCM, and explain the unique focus of each component within the overall enterprise system framework. Source: Unit I | CIE-I, 10/11/2025	10	2	2

ERP, CRM and SCM are the three core enterprise-class applications. EIS includes the internal back-office operations and the external front office (financial system, human resources, inventory, shipping, customer order processing, warehouse).

Basis	ERP	CRM	SCM
Definition	Integrated software that unifies business functions across departments into a single database and user interface.	Strategic approach integrating technology, processes and people to manage interactions with current and potential customers.	Integration and coordination of the flow of goods, services, information and finances from suppliers to the final consumer.
Focus	Internal processes (back office): efficiency and integration	Customers (front office): satisfaction, retention, revenue	Partners across the chain: right product, right time, right cost
Main modules	Finance, HRM, manufacturing, inventory, project management, BI	Operational, analytical, collaborative, strategic and campaign CRM	Planning, sourcing, manufacturing, distribution, returns
Users	All employees	Sales, marketing, service staff	Procurement, logistics, suppliers, distributors
Example	Automatic balance sheet; payroll from attendance	Salesforce tracking leads and follow-ups	Real-time tracking of materials from vendor to warehouse

Within the framework: ERP is the transactional backbone and common database; CRM extends it to the customer side and SCM to the supplier and logistics side. Together they give an end-to-end, demand-to-supply view of the enterprise.

4	Provide a summary of the features of various Enterprise Resource Planning (ERP) modules. Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q2b	08	2	1

ERP software is sold in modules that share a single database. Typical modules (SAP terminology, with equivalents in Oracle and Microsoft Dynamics):

Code	Module	Features
FI	Financial accounting	General ledger, accounts payable and receivable, generation of financial statements
CO	Controlling	Internal management and cost analysis by cost centre, budgeting
AM	Asset management	Fixed-asset purchase, depreciation and asset records
HR	Human resources	Recruitment, hiring, training, payroll, attendance and benefits
MM	Materials management	Purchasing, raw-material inventory, work in progress, finished goods

PP	Production planning	Production planning and scheduling, MRP, actual production
SD	Sales and distribution	Sales orders, scheduled deliveries, customer information, billing
QM	Quality management	Product inspections, material certifications, quality control
PM	Plant maintenance	Preventive maintenance and maintenance resource management
PS	Project system	Project planning, budgeting and tracking
CA	Cross application	Workflow, business information warehouse, e-mail, industry solutions, CRM, SRM

Common features: single integrated database, modular and open architecture (modules can be added or detached), best business practices, real-time reporting, configurability, multi-currency and multi-language support, and on-line connectivity beyond the enterprise.

5	Sketch the demand-supply chain service provider model. Explain the concepts of Enterprise Information Systems (EIS). Source: Unit I | SEE Apr/May 2024 (21IM55B2), Q2a	08	2	2

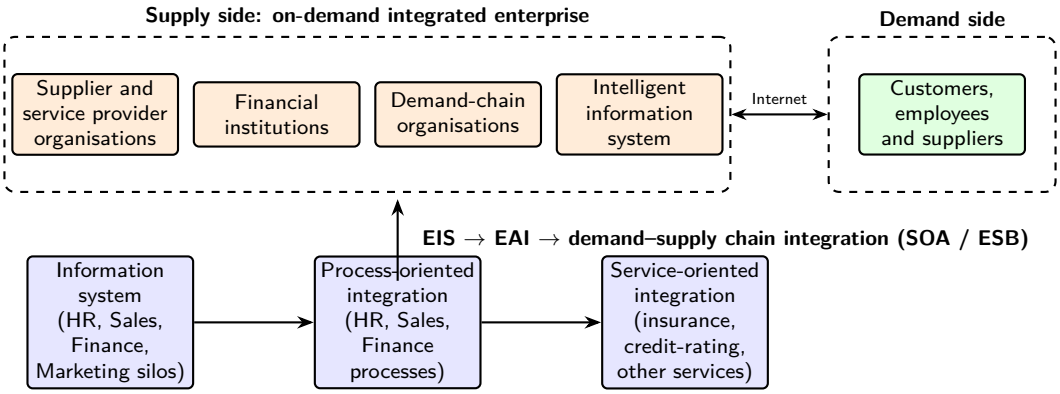


Figure 1: Enterprise demand–supply chain (service provider) model

The model: on the supply side, the on-demand integrated enterprise links supplier and service-provider organizations, financial institutions, demand-chain organizations and an intelligent information system; on the demand side are customers, employees and suppliers, connected through the Internet. Information systems evolve from functional silos (HR, sales, finance, marketing) to process-oriented integration and finally to service-oriented integration (SOA/ESB), where external services such as insurance and credit rating are plugged in.

Concepts of EIS

- **Enterprise-wide integration:** EIS facilitates enterprise-wide integrated information systems covering all functional areas and performs core corporate activities; it uses database systems that are integrated with each other.
- **Applicable to all enterprises:** not only manufacturing; planning, manufacturing, sales and marketing are combined in one business management system.
- **Reengineering of business:** EIS is not merely reengineering systems; it reengineers the way organizations accomplish business tasks, transforming the organization from a functional into a process-centric entity.
- **Planning of the 5 Ms:** EIS plans an enterprise's Man, Money, Material, Method and Machinery to their best synergistic values.
- **Right information to the right person at the right time:** integrated, uniform, up-to-date information gives authority to the right person to make decisions.
- **Features:** best business practices, beyond-the-enterprise connectivity, comprehensiveness, flexibility (client–server, ODBC) and a modular, open architecture.

6	Distinguish between MRP-I and MRP-II. Explain the primary functions of each planning system and illustrate how the evolution from MRP-I to MRP-II broadened the scope of enterprise planning. Source: Unit I | CIE-I, 10/11/2025	10	2	2

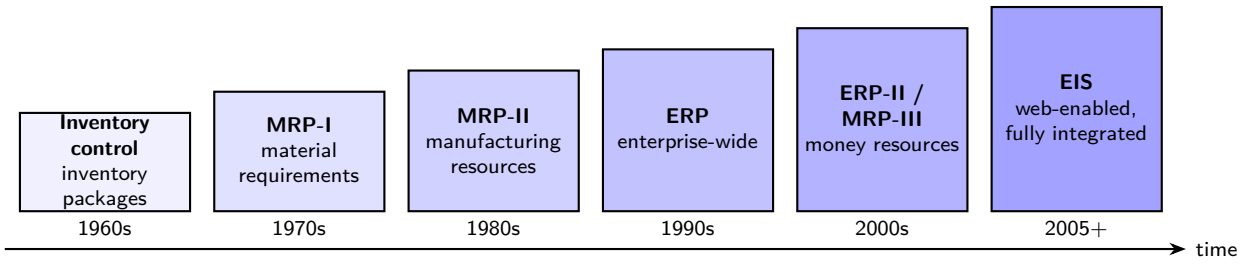


Figure 2: Evolution from inventory control to MRP, MRP-II, ERP and EIS

MRP-I (Material Requirements Planning)

MRP is a time-phased order-release system that schedules and releases manufacturing work orders and purchase orders so that sub-assemblies and components arrive at the assembly station just as they are required. It starts from three documents: the forecast (master production schedule) for end items, the bill of materials and the inventory records, and answers when to order, how much, from whom and when items must be on hand.

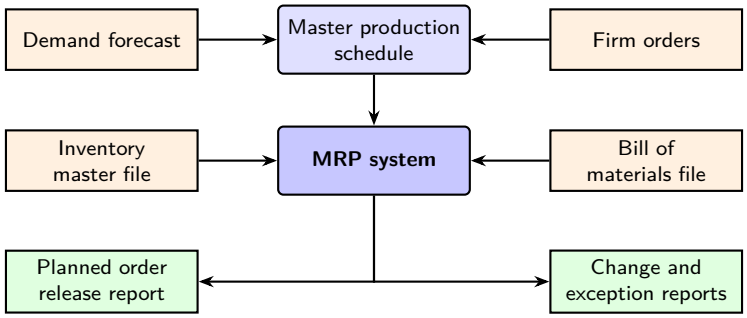


Figure 3: Inputs and outputs of an MRP system

Functions: calculating net requirements and procurement quantities and dates, time-phased planned orders, order release notices, action notices to expedite or cancel, priority and inventory status reports. **Benefits:** reduced inventories, improved customer service and efficiency.

MRP-II (Manufacturing Resource Planning)

MRP-II is a method to plan **all** resources of a manufacturer. Business planning, sales and operations planning, production planning, master scheduling, capacity requirements planning and shop-floor control are tied into the system, which is integrated with accounting and finance to produce business plans, shipping budgets, inventory projections and purchase plans. Its purpose is to integrate production, marketing and finance (and personnel, engineering, purchasing) into the planning process.

Basis	MRP-I	MRP-II
Full form	Material Requirements Planning	Manufacturing Resource Planning
Period	1970s	1980s
Scope	Materials only (dependent-demand items)	All manufacturing resources: materials, machines, labour, money
Capacity	Assumes infinite capacity	Capacity requirements planning and shop-floor control (closed loop)
Integration	Production and purchasing	Production, marketing, finance, engineering, purchasing
Outputs	Planned orders, release and action notices	Business plans, budgets, inventory projections, simulation (“what-if”)

How the evolution broadened enterprise planning

MRP extended inventory control to time-phased material planning; closed-loop MRP added capacity planning and shop-floor feedback; MRP-II brought finance and marketing into the plan, so a single business plan drove production

and money. This laid the foundation for ERP (1990s), which covers all functions of the enterprise (engineering, finance, HR, project management), for MRP-III (money resource planning) and finally for the web-enabled, fully integrated EIS.

7	Evaluate the role of EIS in Customer Relationship Management (CRM) with specific examples of how companies have leveraged EIS for CRM. Source: Unit I | CIE-I, 27/11/2024	10	2	3

CRM integrates technology, processes and people to manage interactions with current and potential customers. EIS provides the integrated customer data, processes and analytics on which CRM depends.

Role of EIS in CRM

(1) **360° customer view:** EIS integrates customer data from sales, service, finance and logistics into a single record, so every touchpoint sees the same history (collaborative CRM).
(2) **Operational CRM:** automation of sales force, marketing and service processes: lead management, order management and service requests.
(3) **Analytical CRM:** data warehousing, data mining and BI on EIS data reveal buying patterns, segments, profitability and lifetime value.
(4) **Campaign management:** targeted, multi-channel campaigns with measurement of effectiveness.
(5) **Integration with ERP/SCM:** order status, availability and delivery dates are available to customer-facing staff (front office linked to back office).
(6) **Strategic CRM:** long-term retention and loyalty programmes aligned with business strategy.

Examples

- **Amazon:** integrates purchase history with AI-based recommendations and its supply chain to personalize offers and give reliable delivery promises.
- **Salesforce users (e.g. Tata Motors dealers):** automated lead tracking and follow-up reminders improve conversion.
- **Banks (HDFC, ICICI):** analytical CRM on transaction data for cross-selling of loans and cards and for churn prediction.
- **Airlines and hotels (Marriott, IndiGo):** loyalty programmes built on integrated guest/passenger data.

Evaluation: EIS-enabled CRM raises customer satisfaction and loyalty, improves sales effectiveness and reduces service cost; its success depends on data quality, integration with back-office systems and user adoption.

8	Describe the hierarchical role of an Enterprise Information System (EIS) in supporting decision-making across the three organizational management levels: strategic, tactical and operational. Source: Unit I | SEE Feb/Mar 2026, Q2b	08	1	2

EIS integrates processes and data into a unified framework and supports managers at all three levels with timely, accurate information.

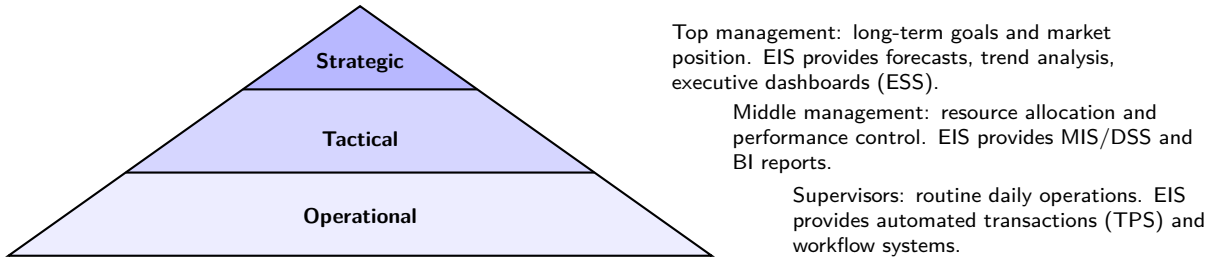


Figure 4: Levels of decision support provided by an EIS

Level	Type of decision	EIS contribution	Example
Strategic (top)	Long-term goals, market positioning; unstructured	Executive support systems, forecasting tools, trend analysis, dashboards	Forecast demand and decide on a new plant or market entry

Tactical (middle)	Resource allocation, performance control; semi-structured	MIS and DSS, BI reports, budget vs actual analysis	Region-wise sales analysis to reallocate the sales force
Operational (lower)	Routine daily operations; structured	Transaction processing, automated workflow, alerts	Automatic purchase order when stock falls below the reorder point

Because all three levels draw on the same integrated database, decisions are consistent: operational transactions feed tactical reports, which roll up into strategic dashboards, and strategic plans cascade down as budgets and schedules.

9	Explain the scope of Digital Transformation in the context of EIS. Describe what the term means for a modern enterprise and how the adoption of an integrated EIS serves as the foundational enabler for achieving digital transformation goals. Source: Unit I | CIE-I, 10/11/2025	10	2	1

Digital transformation (DT) is the strategic adoption of digital technologies to radically improve business processes, culture and customer experiences. It goes beyond digitization: it redefines how organizations create value in a hyper-connected, data-driven world (McKinsey: "the fundamental rewiring of how an organization operates").

Scope of DT for a modern enterprise (core dimensions)

Dimension	Scope
Technology	Integration of cloud computing, AI, IoT, big data and mobile platforms into enterprise systems
Process	Reengineering workflows using automation, RPA and agile methods to improve speed and efficiency
People	Empowering employees with digital tools, a culture of innovation and up-skilling
Strategy	Aligning digital initiatives with business goals for competitive advantage and customer-centricity

Integrated EIS as the foundational enabler

(1) **Single source of truth:** an integrated EIS (ERP, CRM, SCM) removes data silos, which is the prerequisite for analytics and AI.
(2) **Real-time data access** for informed decision-making.
(3) **Cross-functional integration** of processes from order to cash and procure to pay.
(4) **Scalable infrastructure** through cloud-based EIS platforms.
(5) **Enhanced customer engagement** through personalized, omnichannel services (AI in CRM for predictive behaviour analysis; IoT-enabled SCM for real-time inventory).

Outcomes: operational efficiency, agility and innovation, customer-centricity and a data-driven culture. **Challenges:** integration of legacy systems, cybersecurity, change management and data privacy.

10	Explain the concept of Digital Transformation (DT). Describe the crucial role of two key technologies, Cloud Computing and Artificial Intelligence (AI), in driving this transformation within a modern enterprise. Source: Unit I | SEE Feb/Mar 2026, Q2a	08	1	2

Digital transformation is the strategic integration of digital technologies into all areas of an organization to fundamentally change how it operates and delivers value to customers, reimagining processes, culture and customer experience.

Role of cloud computing

- Flexible, scalable and cost-effective deployment of enterprise applications (SaaS ERP and CRM) without heavy capital investment.
- Pay-per-use pricing and rapid provisioning shorten implementation time.
- Anytime, anywhere access supports remote work and global collaboration across the supply chain.
- Provides the storage and computing power needed for big data analytics and AI.
- **Example:** SAP S/4HANA Cloud or Salesforce let a company roll out ERP and CRM across countries within months.

Role of artificial intelligence

- Intelligent automation of tasks (cognitive RPA, invoice reading, chatbots).
- Predictive analytics for demand forecasting, maintenance and fraud detection.
- Personalization of customer experiences (recommendations, next-best offer).
- Natural language processing for customer service and document processing.
- **Example:** Amazon uses AI and analytics in its supply chain and customer service, setting industry benchmarks.

Together, the cloud supplies scalable infrastructure and data, and AI turns that data into decisions and automation, which is the essence of digital transformation.

11	You are advising a high-growth tech startup. The founders are considering which EIS component to prioritize first: CRM for managing investor relations and early customer feedback, or a modular ERP for integrating the finance and HR functions. Argue for the strategic component that should be prioritized to support the startup's growth and eventual digital transformation. Source: Unit I | CIE-I, 10/11/2025	10	3	3

Recommendation: prioritize a modular, cloud-based ERP core (finance first, then HR), with a light CRM added early as a module of the same platform.

Arguments

(1) **Integrated foundation:** ERP provides the single common database and standardized processes on which every other component (including CRM) depends; starting with isolated tools creates data silos that are costly to integrate later.

(2) **Financial control for growth and investors:** a high-growth startup must manage cash burn, billing, payroll, compliance (GST, statutory filings) and produce reliable financial statements for due diligence and funding rounds. Finance modules are also the easiest and most structured to implement.

(3) **Scalability:** as headcount grows rapidly, an HR module handles recruitment, payroll and performance consistently.

(4) **Modularity and cost:** cloud ERP can be started with finance and HR only and extended by modules (phased implementation), keeping cost low.

(5) **Digital transformation path:** clean, integrated data enable analytics and AI later; CRM, SCM and BI plug into the ERP core.

Role of CRM

Investor relations and early customer feedback are important but can initially be handled by a simple CRM (e.g. HubSpot or the CRM module of the same ERP suite) integrated with the finance core. If the startup were purely sales-driven (B2C, many customers), CRM could come first; but for sustainable growth, an integrated ERP core is the strategic priority.

12	Highlight some of the issues you have faced in using SAP implemented in your institution. How did you resolve those issues? Source: Unit I | CIE-I, 27/11/2024	10	3	2

Typical issues faced by students and staff using an SAP-based campus system (student information, fee payment, examination and attendance) and how they were resolved:

Issue	Cause	Resolution
Complex, non-intuitive screens and many steps for simple tasks	Generic SAP GUI; transactions designed for administrators	Used help documents and training videos; requested simplified Fiori/web screens and shortcuts
Slow response or time-outs during peak periods (registration, results)	Many concurrent users, limited server capacity	Avoided peak hours; institution scaled server capacity and staggered schedules
Incorrect or delayed data (attendance, fee receipts, marks)	Manual entry errors, delayed updates by departments	Raised tickets with the help desk; data validated at source by faculty; reconciliation with receipts

Login and password problems	Account locks, single sign-on not configured	Password reset through the IT cell; self-service reset enabled
Limited mobile access	Desktop-oriented interface	Used the web portal or mobile app version
Lack of training and awareness	No orientation for new users	Peer help, user manuals and orientation sessions organized by the institution

Lesson: these are typical EIS implementation issues: user adoption, training, data quality, infrastructure and customization. They are resolved through training, a help desk, data ownership at the source and periodic system improvement.

13	XYZ Manufacturing Company has been facing inefficiencies in its operations due to a lack of integration between its departments. The procurement, inventory, production and sales teams operate on standalone systems, leading to delays, data inaccuracies and increased costs. The company implemented an ERP system, but after six months it is still struggling with system adoption and is not seeing the expected improvements. Analyze the challenges XYZ Manufacturing is facing in implementing the ERP system. Suggest strategies to overcome these challenges and ensure successful adoption. Source: Unit I | CIE-I, 27/11/2024	10	4	4

Challenges faced by XYZ Manufacturing

(1) **Resistance to change:** employees used to standalone systems for years resist new procedures; the cultural aspect of business process reengineering is the biggest hurdle.
(2) **Inadequate training:** training is typically under-budgeted; users do not know how to perform their tasks in the ERP.
(3) **Processes not reengineered:** old, departmental processes were simply automated, so integration benefits were not realized.
(4) **Poor data migration and quality:** inaccurate legacy data carried over, so users distrust ERP reports.
(5) **Weak top-management commitment and project management:** inadequate project definition, planning and change management.
(6) **Excessive customization or misfit:** the software does not match practices, or heavy customization causes errors and delays.
(7) **Parallel use of old systems:** teams continue with spreadsheets, so the ERP is not the single source of truth.

Strategies for successful adoption

(1) **Executive sponsorship and a change-management plan:** communicate the purpose and benefits; appoint process owners and champions in every department.
(2) **Comprehensive, role-based training** with hands-on practice, user manuals and a help desk.
(3) **Reengineer and standardize processes** (procure-to-pay, order-to-cash) and adopt the ERP's best practices; minimize customization.
(4) **Data cleansing** and validation of master data (materials, vendors, customers) before and after migration.
(5) **Retire legacy and standalone tools** after a defined parallel run, so all transactions go through the ERP.
(6) **Phased rollout and quick wins:** stabilize core modules (finance, materials) first; show improvements with KPIs.
(7) **Monitor with KPIs:** order cycle time, inventory accuracy, on-time delivery, user log-ins; gather feedback and improve continuously.
(8) **Vendor and consultant support** for post-implementation optimization.

With these measures the ERP becomes the common platform for procurement, inventory, production and sales, removing delays and data inaccuracies and reducing cost.

14	What is a Business Process? Provide a comprehensive definition, and explain why identifying and defining these processes is the crucial first step before any automation or reengineering effort. Source: Unit II | CIE-I, 10/11/2025	10	2	2

A **business process** is a set of logically related, structured activities or tasks, performed by people or systems, that takes one or more inputs and transforms them into an output of value to an internal or external customer. It has a defined start and end, an owner, inputs, outputs, resources, rules and performance measures, and it usually cuts across functional departments. **Examples:** order-to-cash (order receipt to payment collection), procure-to-pay, hire-to-retire, product development.

Processes are classified as **core (primary) processes** that deliver value to customers (order fulfilment, production), **support processes** (HR, IT, accounting) and **management processes** (planning, governance).

Why identifying and defining processes is the crucial first step

(1) **Understanding the current state (as-is):** mapping shows how work actually flows, the hand-offs, delays, duplications and bottlenecks.
(2) **Avoid automating inefficiency:** automating a poor process only makes mistakes faster; non-value-adding steps must first be removed.
(3) **Set scope and priorities:** defining process boundaries, owners and metrics identifies which processes give the highest benefit from automation or reengineering.
(4) **Establish baseline measures:** cycle time, cost and error rate are measured so that improvement can be demonstrated.
(5) **Requirements for systems:** a defined process (BPMN map) is the specification for configuring ERP or designing RPA bots and workflows.
(6) **Stakeholder alignment:** a shared process definition builds agreement among departments and reduces resistance.

15	Develop the typical flow of inputs converted to outputs through a process. Choose a process and list the alternatives for accomplishing the same. Design the best process. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q3a	08	3	4

Every process converts inputs into outputs through a sequence of activities, with feedback for control.

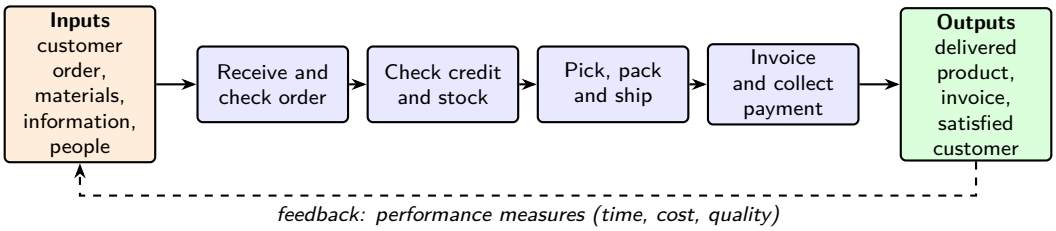


Figure 5: Typical flow of inputs to outputs: order-fulfilment process

Process chosen: customer order fulfilment of a manufacturer

Alternative	Description	Assessment
A. Manual, functional process	Order taken on phone/e-mail; sales, stores, dispatch and accounts each keep their own records; paper hand-offs	Slow (several days), errors from re-entry, no visibility of order status
B. Partly computerized (standalone systems)	Separate sales, inventory and accounting software; data transferred by spreadsheets	Faster, but duplicate entry and data mismatch between departments
C. Outsourced fulfilment (3PL)	Logistics partner stores and ships the goods	Reduces logistics effort but needs strong integration and control
D. Integrated ERP-based process with automation	Online order entry into ERP; automatic credit and stock check; pick list and shipping documents generated; e-invoice and payment tracking	Single entry, real-time status, shortest cycle time, lowest error rate

Best process design (alternative D)

(1) Customer places the order through a web portal or EDI; the order is captured once in the ERP (sales and distribution module).
(2) Automatic credit check and available-to-promise stock check; if stock is short, MRP triggers production or purchase.
(3) Warehouse receives an electronic pick list; barcode scanning confirms picking and packing; shipping documents are generated.
(4) Invoice is generated automatically on dispatch and sent electronically; receipts are matched automatically in accounts receivable.
(5) Dashboards track order cycle time, on-time delivery and errors for continuous improvement.

Benefits: elimination of non-value-adding steps and re-entry, cycle time reduced from days to hours, fewer errors and better customer service.

16	Enumerate the typical infrastructural processes in an organization. Cite examples. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q3b	08	2	2

Infrastructural (support) processes do not directly create the product for the customer but provide the resources and services that core processes need. Typical infrastructural processes:

Process	Purpose	Examples
Human resource management	Recruit, train, pay and develop people	Hire-to-retire, payroll, performance appraisal
Financial management and accounting	Record, control and report money flows	Budgeting, accounts payable/receivable, financial reporting
Information technology management	Provide and maintain IT systems and data	Help desk, network management, ERP support, data backup
Procurement and supplier management	Acquire materials, services and assets	Procure-to-pay, vendor evaluation
Facilities and asset maintenance	Keep buildings, plant and equipment working	Preventive maintenance, utilities, housekeeping
Quality management	Ensure standards and compliance	Audits, calibration, ISO certification
Legal, compliance and risk management	Meet regulations and manage risk	Statutory filings, contracts, safety, cybersecurity
Administration and communication	General services and internal communication	Document management, travel, office services

These processes are prime candidates for shared services, standard ERP modules and automation, because they are similar across organizations.

17	List and explain various methods of business process automation. Source: Unit II | CIE-II, 17/12/2025	10	2	2

Business Process Automation (BPA) uses technology to execute recurring business processes with minimal human intervention. The main methods are:

(1) **Workflow automation:** routing of tasks, documents and approvals according to rules (e.g. leave or purchase-requisition approval).
(2) **Robotic Process Automation (RPA):** software bots mimic human actions on existing applications for repetitive, rule-based tasks (data entry, reconciliation).
(3) **Integration automation (APIs, EDI, middleware):** systems exchange data automatically, e.g. ERP with supplier portals and banks.
(4) **Document and data capture automation:** OCR and intelligent document processing extract data from invoices and forms.
(5) **Decision (rules-based) automation:** business-rules engines take routine decisions, e.g. credit approval within limits.
(6) **Customer interaction automation:** chatbots, IVR, self-service portals and automated notifications.
(7) **Intelligent automation (AI/ML):** prediction and classification added to automation for unstructured or

judgement-based tasks.

(8) **Business Process Management Suites (BPMS) and low-code platforms:** model (BPMN), execute, monitor and improve end-to-end processes.

The method is chosen according to the nature of the task: structured and rule-based tasks suit RPA and workflow; system-to-system data flows suit APIs; unstructured documents need OCR and AI.

18	A manufacturing company's procurement process is complex. Apply three different methods of BPA to the procurement process (from requisition to payment). Justify how each method specifically enhances efficiency in that stage. Source: Unit II | SEE Feb/Mar 2026, Q3a	08	2	3

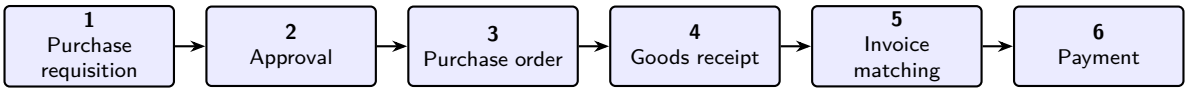


Figure 6: Procure-to-pay process

BPA method	Stage applied	How it enhances efficiency
1. Workflow automation	Requisition and approval	Requisitions are raised on an e-form; rules route them automatically to the right approver by amount and cost centre, with reminders and escalation. Approval time falls from days to hours; full audit trail; no lost paperwork.
2. Integration automation (ERP–supplier portal/EDI)	Purchase order and goods receipt	Approved requisitions are converted into purchase orders in the ERP and transmitted electronically to suppliers; advance shipping notices and barcode scanning at receipt update inventory automatically. Removes re-keying and errors, gives real-time visibility.
3. RPA with OCR (intelligent document processing)	Invoice processing and payment	Bots read supplier invoices (e-mail/PDF), extract data, perform three-way matching (PO, goods receipt, invoice) in the ERP and post invoices; exceptions go to staff; payments are scheduled to capture early-payment discounts. Processing cost and time per invoice fall sharply.

Together the three methods create a touchless procure-to-pay cycle with shorter cycle time, lower cost, better compliance and better supplier relationships.

19	A global bank wants to automate its customer interaction process. Design a BPA solution that combines Customer Interaction Automation and Decision Automation. Detail the steps involved and the types of tools necessary for this solution. Source: Unit II | SEE Feb/Mar 2026, Q3b	08	2	3

Objective: fast, consistent, 24×7 service to customers with automated, rule-based decisions (e.g. account queries, card blocking, small loan or credit-limit requests).

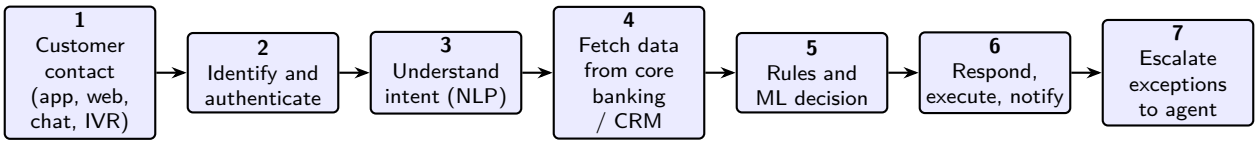


Figure 7: Automated customer interaction and decision process

Steps

(1) **Map and redesign the process:** identify high-volume interactions (balance enquiry, card block, address change, loan eligibility); define decision rules and approval limits.

(2) **Customer interaction automation:** omnichannel front end (mobile app, website chatbot, WhatsApp, IVR) with secure authentication (OTP, biometrics).

(3) **Intent recognition:** an NLP chatbot classifies the request and collects required details.
(4) **Integration:** APIs connect the chatbot to core banking, CRM and credit bureau systems to fetch customer data in real time.
(5) **Decision automation:** a business-rules engine plus ML credit-scoring model decides (e.g. approve a pre-qualified personal loan within limits, flag fraud risk).
(6) **Execution and notification:** RPA bots or APIs update records, generate documents and send SMS/e-mail confirmations.
(7) **Exception handling:** complex or high-risk cases are routed to human agents with the full context (human-in-the-loop).
(8) **Monitor and improve:** dashboards on response time, resolution rate, customer satisfaction and model accuracy; compliance and audit logs.

Tools required

Category	Tools
Customer interaction	Conversational AI/chatbot platforms (e.g. Dialogflow, Microsoft Bot Framework), IVR, mobile and web apps
Integration	APIs, enterprise service bus / middleware, core banking and CRM connectors
Decision automation	Business-rules engine (BRMS), ML credit-scoring and fraud models
Process automation	RPA (UiPath, Automation Anywhere), BPM/workflow suite
Security and analytics	Identity and access management, encryption, BI dashboards, audit logging

20	Design a simplified approach to integrate Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes. Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q4a	10	4	3

Objective: keep the right stock in the right warehouse at all times while minimizing manual work, stock-outs and excess inventory.

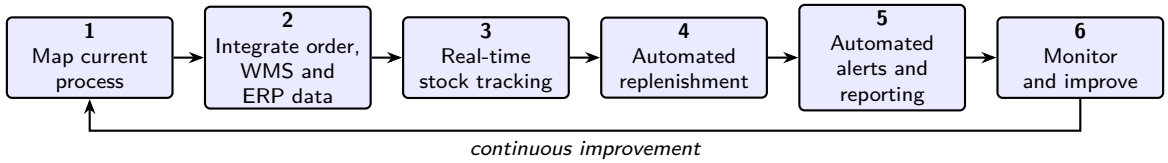


Figure 8: Simplified approach to BPA in e-commerce inventory management

Key steps

(1) **Analyse the as-is process:** map order receipt, stock updates, replenishment and returns; identify manual steps and errors.
(2) **Integrate systems:** connect the e-commerce storefront, warehouse management system (WMS), ERP and suppliers through APIs so that every sale, receipt and return updates stock automatically.
(3) **Real-time tracking:** barcode/RFID scanning at receiving, picking and dispatch; a single inventory record across warehouses.
(4) **Automated replenishment:** reorder points and demand forecasts (from sales history and seasonality) trigger purchase orders to suppliers automatically.
(5) **Automated order allocation:** orders are routed to the nearest warehouse with stock.
(6) **Alerts and reporting:** low-stock, slow-moving and expiry alerts; dashboards on stock levels and turnover.
(7) **Returns automation:** returned items are inspected, graded and restocked or written off through workflow.
(8) **Pilot, train and monitor:** pilot on one category, train staff, track KPIs and improve.

Expected outcomes

- Real-time, accurate inventory visibility across channels and warehouses.
- Fewer stock-outs and overselling; lower excess and dead stock.

- Faster order fulfilment and higher customer satisfaction.
- Reduced manual effort and errors; lower inventory carrying cost.
- Data for better forecasting and purchasing decisions.

21	Define Robotic Process Automation (RPA). Explain how it works. List its uses and benefits. Source: Unit II | CIE-II, 17/12/2025	10	2	2

Robotic Process Automation (RPA) is a software technology that uses software robots (bots) to emulate the actions of a human interacting with digital systems: logging into applications, reading and entering data, copying between systems, filling forms, calculations and triggering responses, to execute high-volume, repetitive, rule-based tasks.

How RPA works

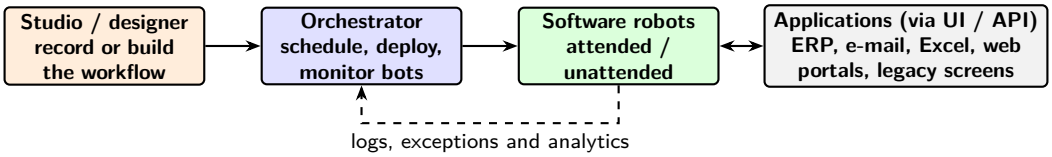


Figure 9: Components of an RPA system

(1) **Process selection:** choose a rule-based, repetitive, stable process with structured digital inputs.
(2) **Design/recording:** in the RPA studio the developer records or builds the workflow (screen actions, rules, exceptions), often with drag-and-drop activities.
(3) **Deployment:** bots are deployed through the orchestrator (control room), which schedules and assigns work, manages credentials and queues.
(4) **Execution:** bots work through the user interface (or APIs) of existing applications, like a human user, but faster and without errors; attended bots assist a person, unattended bots run on their own.
(5) **Monitoring:** logs, exception handling and analytics; exceptions are passed to humans.

Uses

- Finance: invoice processing, bank reconciliation, report generation.
- HR: onboarding, payroll inputs, employee data updates.
- Supply chain: order entry, purchase-order creation, shipment tracking updates.
- Customer service: account updates, claim processing, data validation.
- IT: user account creation, password resets, system monitoring.

Benefits

- Speed and 24×7 operation; shorter cycle time.
- Accuracy and consistency; fewer errors.
- Cost savings; quick return on investment.
- Non-invasive: works with existing systems without changing them.
- Scalability: bots can be added during peaks.
- Compliance: complete audit trail.
- Employees freed for higher-value work.

22	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA). Discuss how implementing RPA can streamline the process and improve efficiency. Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q3a	10	3	3

Current problem

Orders arrive by e-mail, PDF and spreadsheet; staff manually key them into the ERP, validate customer, price, stock and credit details and send confirmations. This causes delays, errors and backlogs at peak times.

Proposed RPA solution

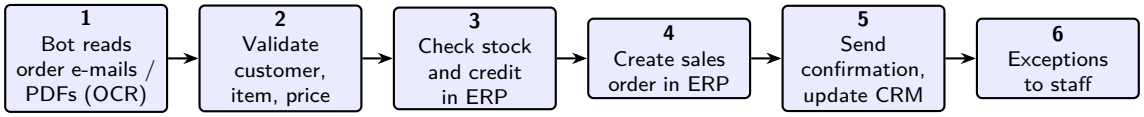


Figure 10: RPA-enabled order processing

(1) **Order capture:** an unattended bot monitors the order mailbox and portal, extracts order data using OCR/intelligent document processing.
(2) **Validation:** the bot checks customer codes, item numbers, prices and quantities against ERP master data and business rules.
(3) **Availability and credit check:** queries stock and credit limit in the ERP.
(4) **Order entry:** creates the sales order in the ERP, triggering production planning or picking.
(5) **Confirmation:** sends an order acknowledgement with expected delivery date and updates the CRM.
(6) **Exception handling:** mismatches (unknown item, credit hold) are routed to a staff member with details (human-in-the-loop).
(7) **Reporting:** dashboards on orders processed, cycle time and exceptions.

How RPA streamlines the process

- Order processing time reduced from hours to minutes; bots work 24×7 and absorb peaks.
- Data-entry errors eliminated; consistent rule-based validation.
- Faster order-to-cash cycle and improved customer satisfaction.
- Staff redeployed to customer service and exception handling.
- Audit trail and real-time visibility of order status.

Implementation: start with a pilot on the highest-volume order format, measure benefits, then scale; keep the process documented and maintain bots when ERP screens change.

23	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenario. Source: Unit II | Improvement CIE, 29/01/2025; SEE Feb/Mar 2025, Q3b	10	3	3

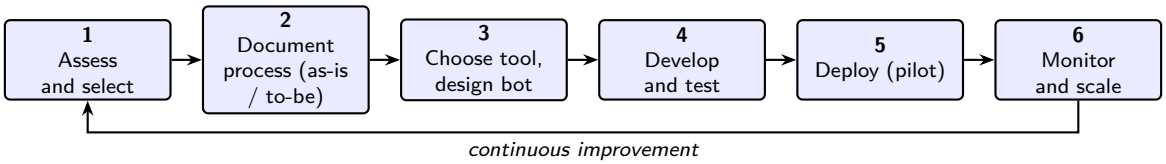


Figure 11: RPA implementation life cycle

(1) **Assess feasibility and select the process:** invoice processing is high-volume, rule-based and uses digital invoices, making it suitable. Estimate volume, time per invoice and expected savings (business case).
(2) **Document the process:** map the as-is steps (receive invoice, extract data, match with PO and goods receipt, approve, post in ERP, schedule payment) in a process definition document; standardize the to-be process and exception rules.
(3) **Select the RPA tool and design the solution:** e.g. UiPath or Automation Anywhere with OCR/intelligent document processing; design bot workflows, data validation rules and exception queues.

(4) **Develop the bots:** configure invoice capture from e-mail/supplier portal, data extraction (vendor, invoice number, date, amount, GST), three-way matching in the ERP, posting and payment scheduling.
(5) **Test:** unit, integration and user acceptance testing with sample invoices including exceptions (missing PO, price mismatch, duplicates).
(6) **Deploy:** pilot with selected vendors; schedule bots in the orchestrator; train finance staff to handle exceptions.
(7) **Monitor and maintain:** track invoices processed, straight-through rate, cycle time, errors; maintain bots when invoice formats or ERP screens change.
(8) **Scale:** extend to all vendors and to related processes (payment reconciliation, vendor statement matching).

Outcome: faster invoice processing, fewer errors and late-payment penalties, capture of early-payment discounts and a full audit trail.

24	Explain the role of RPA frameworks such as BPMN and Agile in the lifecycle of an RPA project. Why are these structured approaches necessary even when dealing with tactical automation? Source: Unit II | SEE Feb/Mar 2026, Q4a	08	2	2

BPMN in the RPA life cycle

Business Process Model and Notation (BPMN) is a standard graphical notation (events, activities, gateways, swim lanes) for modelling processes. In an RPA project it is used to:

- Document the as-is process and design the to-be automated process precisely, including decision points and exceptions.
- Identify which steps are done by bots and which by humans (swim lanes for bot and human).
- Serve as the common language between business users, analysts and RPA developers, and as the basis for the process definition document.
- Support process optimization before automation, and governance and maintenance afterwards.

Agile in the RPA life cycle

Agile delivers automation in short iterations (sprints) with frequent feedback:

- The automation backlog is prioritized by business value; bots are built incrementally (a minimum viable bot first).
- Continuous user feedback and testing in each sprint catch changes in rules or screens early.
- Quick delivery of benefits and easy adaptation to changing requirements.

Why structured approaches are needed even for tactical automation

(1) **Avoid automating bad processes:** modelling first reveals inefficiencies that should be removed.
(2) **Reliability and exception handling:** bots fail when undocumented variations occur; complete process models reduce breakdowns.
(3) **Maintainability:** documented bots can be updated when applications change, and are not dependent on one developer.
(4) **Governance and compliance:** audit trails, security of credentials and controlled change management.
(5) **Scalability:** a consistent framework allows many small automations to grow into an enterprise automation programme (centre of excellence) without "bot sprawl".

25	Explain the defining difference between Basic RPA and Advanced RPA (Cognitive RPA). Discuss how Cognitive RPA, through its integration with AI/ML, enables organizations to handle complex tasks that Basic RPA cannot. Source: Unit II | SEE Feb/Mar 2026, Q4b	08	2	2

Basis	Basic (traditional) RPA	Advanced (cognitive) RPA
Definition	Bots that follow predefined rules to mimic human actions	RPA combined with AI/ML, NLP and computer vision
Data handled	Structured, digital data (forms, spreadsheets)	Unstructured and semi-structured data (e-mails, scanned documents, images, speech)
Decision ability	Only rule-based; "if–then" logic	Judgement-based: classifies, predicts, learns from data
Adaptability	Breaks when inputs or screens change	Learns and improves with more data
Examples	Copying data between systems, report generation	Reading varied invoice formats, sentiment analysis of complaints, fraud detection

How cognitive RPA handles complex tasks

- **Intelligent document processing:** OCR plus ML extracts data from invoices, claims and contracts in any layout.
- **Natural language processing:** understands e-mails and chat messages, identifies intent and sentiment, and routes or answers them.
- **Machine-learning decisions:** predicts outcomes such as credit risk, fraud or demand and takes or recommends decisions.
- **Computer vision:** reads images and screens, verifies documents (e.g. KYC).
- **Continuous learning:** human corrections of exceptions are fed back to improve accuracy (human-in-the-loop).

Example: an insurer uses cognitive RPA to read claim forms and photographs, assess damage with an image model, check policy rules and approve simple claims automatically, which basic RPA could not do because the inputs are unstructured and require judgement.

26	Discuss the importance of incorporating best practices in Business Process Reengineering (BPR) initiatives. Explain how organizations can identify and implement best practices to ensure successful BPR outcomes. Provide examples of best practices commonly used in BPR projects and their impact on business performance. Source: Unit II | SEE Feb/Mar 2025, Q4b	08	2	3

Business Process Reengineering (BPR) is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in cost, quality, service and speed. Because BPR is risky, incorporating proven best practices raises the chance of success.

Importance of best practices

- Reduce the high failure rate of BPR projects by learning from successful organizations.
- Provide proven, benchmarked process designs instead of trial and error.
- Speed up redesign and implementation (e.g. best practices embedded in ERP packages).
- Give measurable targets for performance improvement.

How to identify and implement best practices

(1) **Benchmarking:** compare processes and metrics with industry leaders and other units.
(2) **Reference models:** use ERP best-practice processes, industry frameworks (APQC process classification, SCOR) and consultants' experience.
(3) **Process analysis:** map as-is processes, find gaps against best practice, design the to-be process.
(4) **Pilot and adapt:** test the new design, adapt it to the organization's context, then roll out.
(5) **Change management and training:** communicate, involve employees and train them.
(6) **Measure and sustain:** track KPIs and continuously improve.

Common BPR best practices and their impact

Best practice	Impact on business performance
Organize around outcomes, not tasks (case worker / case team)	Fewer hand-offs; cycle time cut sharply (e.g. IBM Credit reduced financing approval from days to hours)
Capture information once, at the source	Fewer errors and duplicate entries
Have those who use the output perform the process	Fewer co-ordination delays; greater ownership
Treat geographically dispersed resources as centralized (shared databases, IT)	Economies of scale with local flexibility
Link parallel activities instead of integrating their results later	Faster development and fewer rework loops
Put the decision point where the work is performed; build control into the process	Faster decisions, empowered employees
Eliminate non-value-adding steps (e.g. Ford's invoiceless payment)	Large cost savings (Ford reduced accounts-payable staff by about 75%)
Use IT as an enabler, strong leadership and cross-functional teams	Successful, sustained redesign

27	Identify the common myths about reengineering. Elucidate the conditions under which reengineering works best. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q4a	08	3	3

Common myths about reengineering

(1) **Reengineering is the same as automation:** automating existing processes only speeds up inefficient work; reengineering redesigns the process first.
(2) **Reengineering means downsizing:** the aim is dramatic performance improvement; job cuts may follow but are not the objective.
(3) **Reengineering is restructuring or reorganizing:** changing the organization chart does not change how work flows across functions.
(4) **Reengineering is the same as TQM / incremental improvement:** TQM improves existing processes continuously, reengineering replaces them radically.
(5) **Reengineering is an IT project:** IT is an enabler, but the change is in processes, people and culture.
(6) **It is quick and low risk:** reengineering takes time, investment and carries high risk of failure.
(7) **It is only for companies in crisis:** healthy companies also reengineer to stay ahead.

Conditions under which reengineering works best

(1) **Strong top-management leadership and commitment** with a clear vision and stretch goals.
(2) **Focus on a few critical core processes** with high impact on customers (e.g. order fulfilment).
(3) **Clear customer focus:** design starts with what the customer wants.
(4) **Cross-functional teams** with process owners, empowered to redesign.
(5) **IT as an enabler** (integrated databases, ERP, workflow).
(6) **Effective change management:** communication, training and attention to the human and cultural side.
(7) **Willingness to take risks and invest**, with performance measures to track results.
(8) **A real need for dramatic improvement**, where incremental change would not suffice.

28	Provide a comparative analysis of clean-slate and technology-enabled reengineering. Source: Unit II | SEE Apr/May 2024 (21IM55B2), Q4b	08	3	3

Basis	**Clean-slate reengineering**	**Technology-enabled reengineering**
Meaning	Redesign the process from scratch, ignoring the existing process ("start with a blank sheet")	Redesign the process around the capabilities of a technology, usually an ERP or other enterprise package and its best practices
Starting point	Customer needs and ideal process	Process templates available in the chosen software
Degree of change	Most radical; highest potential gains	Substantial but bounded by the software's design
Risk and cost	High risk, high cost, long duration; disruptive	Lower risk; proven processes; faster to implement
Role of IT	IT chosen or built after the process is designed	IT drives and constrains the design
Customization	Systems often custom-built to fit the new process	Minimal customization; organization adapts to the package
Suitability	When existing processes are obsolete or a breakthrough is needed	When standard processes (finance, procurement) are adequate and quick results are needed
Example	Ford's invoiceless accounts-payable process	Redesigning procure-to-pay around SAP's standard MM and FI processes

Analysis: clean-slate reengineering offers the greatest improvement but its risk and cost are high and many projects fail; technology-enabled reengineering is more practical and is now the common approach, though it may lead to processes that are no different from competitors'. Many organizations combine them: clean-slate design for a few distinctive core processes and package-enabled design for standard support processes.

* * *